

Sowmya Sivala

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SUMMARY

Data Analyst with 3+ years of experience transforming complex operational and transactional data into actionable business insights. Applies Python, SQL, and PySpark to consolidate large datasets, improve data quality, and support analytical workflows, while leveraging Snowflake, PostgreSQL, and AWS Athena to enable scalable reporting. Builds Tableau and Power BI dashboards that enhance performance visibility, uses statistical analysis and A/B testing to evaluate outcomes, and delivers data-driven recommendations that strengthen decision-making, process efficiency, and operational oversight.

TECHNICAL SKILLS

Data Analysis & Statistics: Data Analysis, Statistical Analysis, Hypothesis Testing, A/B Testing, Exploratory Data Analysis (EDA), Data Validation

Programming & Querying: Python (Pandas, NumPy), SQL (Joins, CTEs, Window Functions), Excel (Pivot Tables, Power Query)

Data Visualization & BI: Tableau, Power BI, Looker, Data Storytelling, Dashboard Development, KPI Reporting

Data Engineering & Processing: PySpark, ETL, Data Wrangling, Data Cleaning, Data Transformation, Data Modeling

Databases & Cloud Platforms: MySQL, PostgreSQL, Snowflake (Data Warehousing, Data Sharing), AWS (S3, Redshift, Athena)

Analytics & Business Intelligence: Business Analytics, Predictive Analytics, Forecasting, Customer Analytics, Performance Metrics, Root Cause Analysis

PROFESSIONAL EXPERIENCE

Data Analyst, MOHELA

Jan 2025 - Present | Remote, USA

- Orchestrate a federal student loan repayment analytics initiative using Python and Snowflake, consolidating borrower servicing records across programs while improving reporting consistency enterprise-wide.
- Examine repayment behavior trends through SQL-driven segmentation models, uncovering patterns across 18% more borrower cohorts that support targeted outreach strategies and operational planning.
- Construct Tableau dashboards tracking delinquency, enrollment, and repayment milestones, enabling servicing stakeholders to review portfolio performance through standardized metrics and governance controls.
- Reconcile borrower datasets from servicing platforms and regulatory submissions, reducing validation exceptions by 22% through automated quality checks supporting compliance reporting requirements.
- Quantify repayment program effectiveness using Statistical Analysis and A/B Testing frameworks, helping business teams evaluate intervention outcomes across evolving federal servicing initiatives.
- Streamline monthly portfolio reporting workflows through AWS Athena integrations, shortening processing cycles by 31% while maintaining traceability for borrower performance and servicing operations.

Data Analyst, Laurus Labs Limited

Oct 2021 - Jul 2023 | Remote, India

- Investigated manufacturing quality and production yield performance using Power BI and PostgreSQL, consolidating batch records across pharmaceutical operations for centralized analytical visibility.
- Mapped production deviations against process parameters through structured SQL analysis, identifying relationships that lowered recurring investigation workloads by 16% across manufacturing teams.
- Aggregated laboratory, equipment, and batch execution datasets into PySpark pipelines, supporting timely review of manufacturing performance throughout regulated production environments.
- Evaluated process stability using Statistical Validation techniques, improving batch release decision accuracy by 13% while strengthening adherence to pharmaceutical quality expectations.
- Automated recurring KPI reporting for plant leadership through Excel Power Query workflows, eliminating manual consolidation efforts and improving reporting turnaround during audits.
- Correlated production throughput and equipment utilization metrics across facilities, uncovering opportunities that increased manufacturing efficiency by 11% without disrupting validated operating procedures.

CERTIFICATIONS

- Google Advanced Data Analytics Professional Certificate ([Link](#))
- Databases and SQL for Data Science ([Link](#))
- Google Advanced Data Analytics Capstone ([Link](#))
- Go Beyond the Numbers: Translate Data into Insights ([Link](#))

PROJECTS

Customer Churn Analysis | Python, Scikit-Learn, Power BI

- Analyzed 7,032 telecom customer records using Pandas and SQL-style transformations, uncovering contract and service patterns driving customer churn behavior.
- Developed Logistic Regression and Random Forest models, achieving 79% accuracy and 52% churn recall while optimizing retention-focused classification decisions.
- Engineered feature pipelines and evaluated model performance through cross-validation, improving prediction reliability across multiple customer risk segments.
- Built interactive Power BI dashboards with 4 KPI cards and 5 visualizations, highlighting \$139K monthly revenue loss trends.

Patient Health Risk Prediction | Python, XGBoost, SHAP

- Built an end-to-end healthcare risk prediction pipeline using XGBoost, achieving ROC-AUC above 0.98 on Electronic Health Record datasets.
- Processed and integrated patient, observation, and condition datasets using Pandas, streamlining multi-table healthcare data preparation workflows efficiently.
- Applied SMOTENC and feature engineering techniques, improving minority-class representation and enhancing predictive performance for high-risk patient identification.
- Developed a Streamlit application with SHAP-based explanations, delivering personalized risk insights and supporting transparent clinical decision-making processes.

EDUCATION

Master of Science , <i>Saint Louis University</i> Data Analytics GPA: 3.68/4.0	Aug 2023 - May 2025 MO, USA
Bachelor of Science , <i>Sri Sathya Sai Institute of Higher Learning</i> Mathematics and Statistics GPA: 3.01/4.0	Jun 2019 - Apr 2022 AP, India